CO-4 (AT-1)

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Q1) Comparative Analysis

Q1) Compare Hashing and BST Based on Time Complexity and methods.

Q2) Compare The Time Complexity of all notation in Bubble, insertion, selection, Quick, merge, Heap.

1) Comparative Analysis : Hashing vs BS

Feature	Hashing	Binary Search
1. method	uses a hash function to find location of element	Repeatedly divides a sorted array into two halves
2. Data required	Does not require sorted data	Data must be sorted.
3. Search - Best case	O(1)	O(1)
4. Search - Avg case	O(1)	O(log n)
5. Search - Worst case	O(n)	O(log n)
6. Insertion	Avg O(1)	O(n) Sorted order
7. Deletion	Avg O(1)	O(n) Sorted order
8. Advantage	very fast	Efficient Searching
9. Disadvantage	Hash collision	Requires sorted data

2) Time Complexity Comparison of Sorting Algorithms

Sorting Algorithm	Best Case	Avg Case	Worst Case	Space
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$
Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$